isonavi

Autonomous underwater reconnaissance for post-flood disaster response

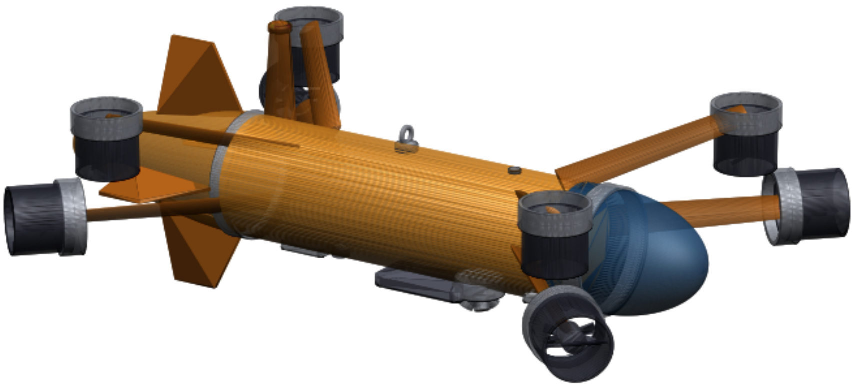
Technical brief | `github.com/AUV-VITP/isonavi`

The problem

When a bridge fails in monsoon flood, the two questions that decide the response are where the casualties are and whether the remaining foundations will hold. Both are answered underwater, and in Indian flood conditions that water is opaque with sediment and running at $2.4\,\mathrm{m\,s^{-1}}$. Divers cannot work in it. Optical cameras see nothing. The reconnaissance simply does not happen, and decisions are made without it.

What has been built

A complete autonomy stack for a vehicle designed for exactly those conditions, with the autonomy validated on flight hardware, the perception validated on real sonar, and the vehicle specified in verified parametric CAD.



Station keeping envelope	$2.69\,\mathrm{m\,s^{-1}}$	Navigation error, no external fix	0.322 m mean of 5
Autonomous mission flown	573 m, 14.9 min	Bathymetric map error	0.285 m
On a RISC-V flight computer	0.107 m	Endurance, slack to design current	17.8 to 1.2 h
Depth rating	50 m	Parts cost, one airframe	₹38.3 lakh

Why the usual platform will not do this

An open frame vehicle of the class normally proposed for this role holds station only to $0.96\,\mathrm{m\,s^{-1}}$. At the design current it is swept away and no controller can prevent it. The purpose designed faired hull raises that to $2.69\,\mathrm{m\,s^{-1}}$ on drag derived from its own geometry.

Comparable imported survey vehicles do not close this gap either, for a structural reason rather than a performance one: REMUS 100 and Iver3 are transit survey torpedoes, a single propeller with control fins, which needs forward speed for control authority and gives no lateral thrust. That is right for long survey lines in open water. It cannot hold station over a scoured pier and orbit it to measure the hole. isonavi gives up depth, endurance and half its drag budget to buy exactly that.

Technology readiness, per subsystem

Subsystem	**TRL**	**Evidence**
Autonomy: estimation, control, mission logic	4	Runs on a RISC-V flight computer inside a 50 ms real-time budget for a complete 894 s mission, driving a separate microcontroller over a CRC checked link.
Actuator interface	4	Eight hardware PWM channels generated on the target microcontroller from the allocation's own output; 17,878 of 17,879 echoes exact, measured from the duty registers.
Acoustic perception	4	Trained and evaluated on real sonar, not simulation. 99.0 % on a held out real test split.
Geometric detection, mapping	3	Validated against simulated ground truth only.
Vehicle structure, hydrodynamics	3	Verified parametric CAD, closed mass and stability budget, closed form depth and drag analysis. Nothing has been in water.

What was actually demonstrated

The autonomy runs on flight hardware. The complete 894 s mission was executed with estimation, control and the mission state machine on a RISC-V single board computer, driven by simulated physics on a host that never gives the board ground truth. It reached the end state with a navigation error of 0.107 m against 0.15 m in pure simulation, using 34.6 ms of its 50 ms budget per tick.

The host and the board agree on the trajectory to roughly one part in 10^{12} over 17,884 ticks. That is the signature of the same arithmetic in the same order, not of two implementations that happen to behave alike.

The allocation reaches an actuator. Over the same mission the flight computer commanded 17,884 pulse width frames to an ESP32 generating eight real 50 Hz channels. 17,878 echoes exactly reproduced a commanded width, read back from the timer's duty registers. 1 frame did not. Zero CRC errors on either link.

Submerged vehicles are found without training data. All 2 large vehicles and 1 small vehicle in the modelled scene are located in every run, at 1.80 m mean error, from bathymetry alone. That reduces an unsearchable 1.54 ha box to roughly 24.4 georeferenced contacts to dive.

What is not claimed

- **No physical vehicle exists.** Hydrodynamic coefficients are closed form estimates cross checked against the CAD geometry, agreeing to 21 % in axial drag. CFD, tow tank and a pressure test remain required.
- **The detector is a screening tool, not a classifier.** It raises about 10.0 spurious contacts per hectare alongside the real ones, and misses a flat deck slab in two runs of 5.
- **Not single fault tolerant for station keeping.** Losing one of four horizontal thrusters halves surge and drops the holdable current to 1.86 $\mathrm{m\,s^{-1}}$. The remedy is priced: 196 N thrusters instead of 120 N.
- **Simulator pretraining does not transfer to real sonar.** Measured and reported as a negative result rather than omitted.

Where the money goes, and why that matters

One airframe costs ₹3,827,800 in parts, about ₹38.3 lakh at ₹95.64 per USD. The distribution is the interesting part: the two acoustic instruments are 59 % of it, and the hull, propulsion, power and the entire autonomy stack together come to ₹1,560,175.

The work in this proposal therefore runs on hardware costing a few tens of thousand rupees, and it is the part that is genuinely ours. Hulls, thrusters and pressure housings are not difficult to buy or to copy; the autonomy that flies a survey in zero visibility against a 2.4 $\mathrm{m\,s^{-1}}$ current is. That is where the effort has gone, and it is why the readiness table above is strongest exactly where the value is.

What the funding buys

Not more analysis. Water. The programme is ₹7,278,908, about ₹72.8 lakh, of which the airframe is ₹3,827,800. The balance is ₹1,140,890 of equipment bought once and used for every build after it, ₹172,248 of stock, and ₹1,358,088 of trials, with 12 % contingency. The full sheet in the technical report gives every line, why it exists and whether its price is a published one, a quotation or a workshop estimate.

Phase	Work	Evidence produced
Months 1 to 6	CFD and tow tank validation; pressure test of the hull section; indigenous thruster and housing sourcing	Measured coefficients replacing the closed form estimates; a qualified depth rating
Months 7 to 12	Integrated vehicle; tank and reservoir trials; sonar and doppler integration	Wet testing against surveyed targets
Months 13 to 24	Instrumented river trials with an operational partner; scour measurement against diver survey	Field data at operational current and turbidity

This is the transition from readiness level 4 to levels 5 and 6, and it is the step where testing facilities and an operational partner change the timeline more than funding alone would. A partner with a tank, a reservoir and an instrumented river reach compresses months 7 to 24 the most.

Every number in this brief is generated from the measured result files by a script in the repository, and is the same value the full technical report carries. The report, the simulator, the autonomy stack, the CAD and the hardware bench are all in one repository, and every result reproduces from a single command.